

H2 SF2524

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Exercise 1

1.a

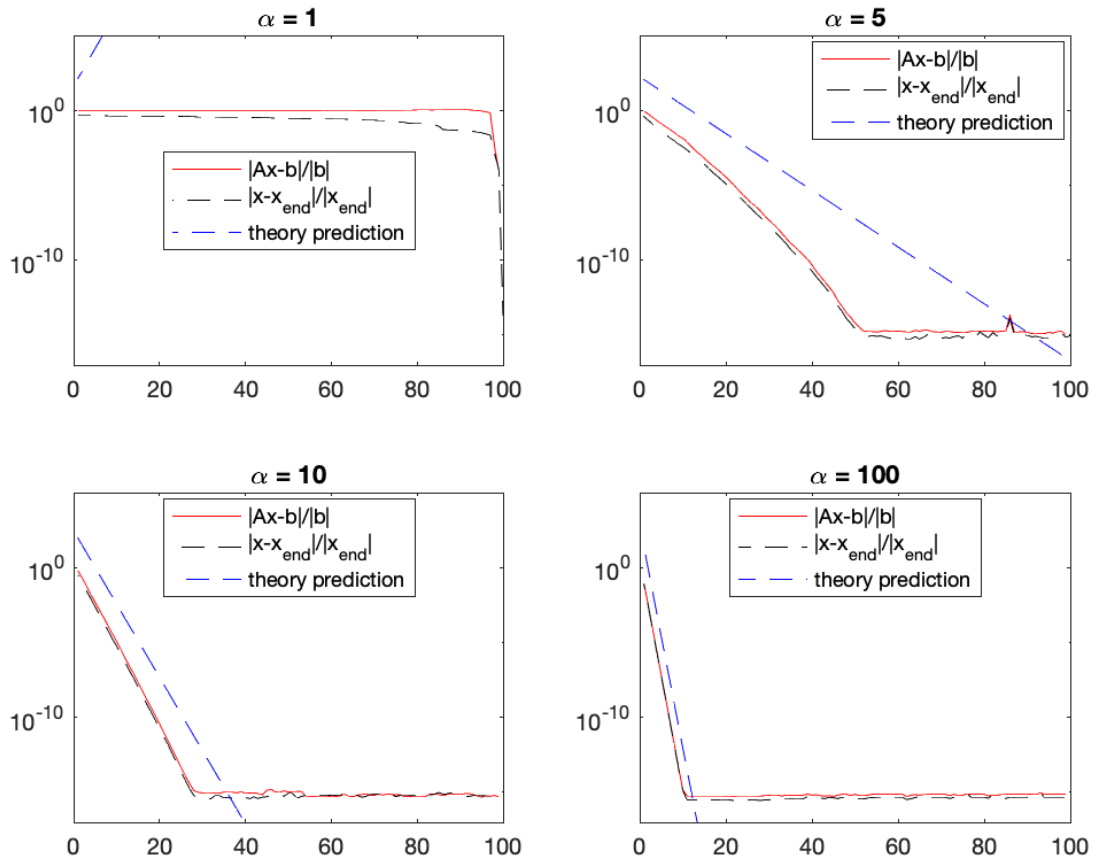


Figure 1: The different errors given by an α with respect to the number of iterations. Observe here the blue theory prediction line represent the theoretically upper-bound for GMRES

1.b

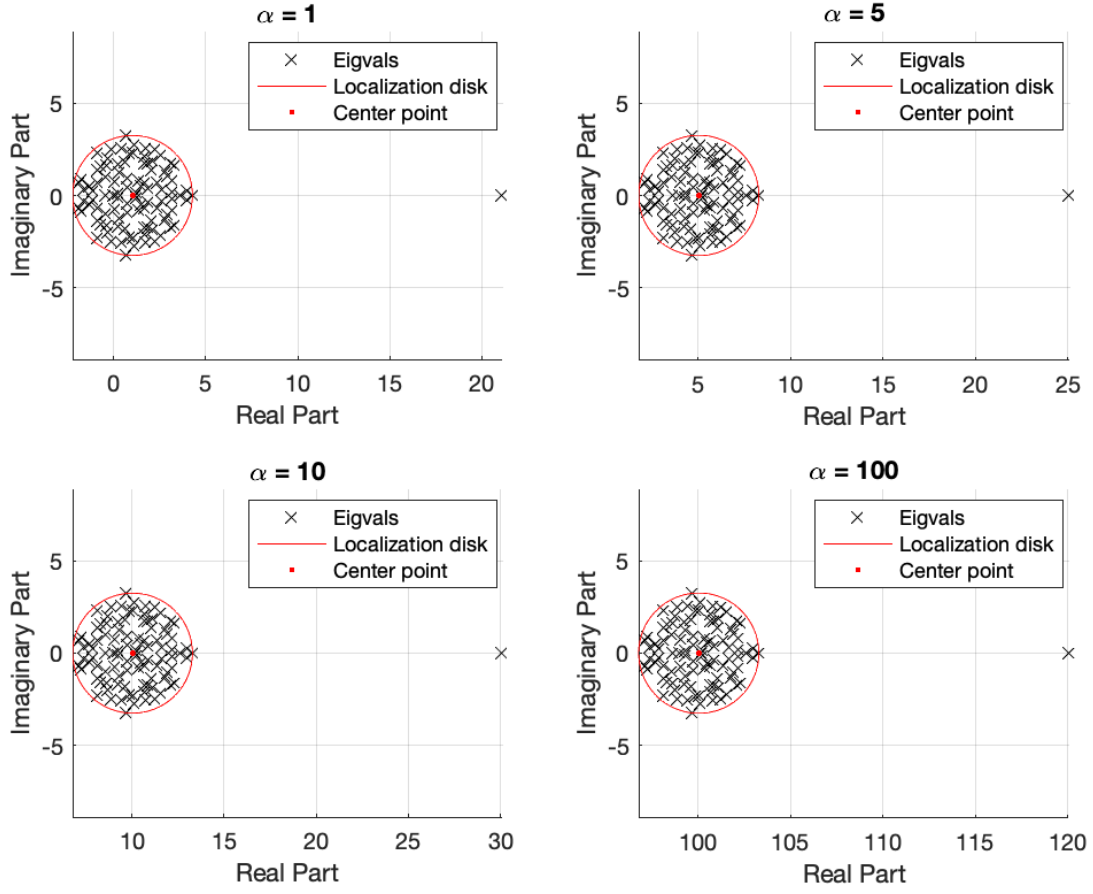


Figure 2: The eigenvalues of $A + \alpha I$ for different α

We can see that we have faster convergence of the method when α is bigger because the eigenvalues are further from the origin (and from the theory we know that

$$\left(\frac{\|Ax_m - b\|}{\|b\|}\right) \leq \|V\| \|V^{-1}\| \left(\frac{\|\lambda - c\| + \rho}{\|\lambda\|}\right) \left(\frac{\rho}{\|c\|}\right)^{m-1}$$

where c is the center of the circle that includes all the eigenvalues except for the only outlier λ we are considering, ρ is the radius of this circle, V is the matrix whose columns are the eigenvectors of A , m is the index of the iterations).

From Figure 2 we see that the eigenvalues are always distributed with the same pattern for each value of α , just translated to the right by α (so in all the four situations the radius of the circle remains the same, but the distance between the center and the origin increases as α gets bigger).

The reason why this happens:

For $A + \alpha I$ the eigenvalue equation is:

$$(A + \alpha I)v = \mu v$$

where $v \neq 0$ is the eigenvector associated to the eigenvalue μ . Then expanding the equation:

$$Av + \alpha Iv = \mu v$$

Since $Iv = v$ this simplifies to:

$$Av = (\mu - \alpha)v$$

This last equation shows that $\mu - \alpha$ are the eigenvalues of A associated to the eigenvector v

Then we proved that the eigenvalues of $A + \alpha I$ are equal to the eigenvalues of A shifted by α

1.c

m	$n = 200$		$n = 500$		$n = 1000$	
	resnorm	time	resnorm	time	resnorm	time
5	3.994386×10^0	0.0014	6.488966×10^0	0.0016	9.087477×10^0	0.0042
10	3.976075×10^0	0.0029	6.480431×10^0	0.0029	9.072191×10^0	0.0061
20	3.887606×10^0	0.0025	6.400149×10^0	0.0075	9.057658×10^0	0.0192
50	3.588471×10^0	0.0146	6.144359×10^0	0.1178	8.874867×10^0	0.0462
100	2.851750×10^0	0.1647	5.836009×10^0	0.0803	8.737229×10^0	0.1339

Table 1: Results for GMRES $\alpha = 1$.

m	$n = 200$		$n = 500$		$n = 1000$	
	resnorm	time	resnorm	time	resnorm	time
5	3.824208×10^{-6}	0.0004	6.808373×10^{-5}	0.0009	4.961989×10^{-4}	0.0031
10	5.492900×10^{-13}	0.0007	9.856985×10^{-11}	0.0016	4.097048×10^{-9}	0.0064
20	5.252644×10^{-15}	0.0021	8.334466×10^{-15}	0.0040	1.849158×10^{-14}	0.0184
50	5.220752×10^{-15}	0.0432	8.425302×10^{-15}	0.0495	1.841472×10^{-14}	0.0729
100	5.220752×10^{-15}	0.1394	1.017375×10^{-14}	0.3289	1.849627×10^{-14}	0.4009

Table 2: Results for GMRES $\alpha = 100$.

α	$n = 200$		$n = 500$		$n = 1000$	
	Time	Resnorm	Time	Resnorm	Time	Resnorm
1	0.0870	2.938864×10^{-14}	0.0452	6.272661×10^{-14}	0.1807	1.474944×10^{-12}
100	0.0030	3.453342×10^{-15}	0.0299	7.294742×10^{-15}	0.1319	1.362950×10^{-14}

Table 3: Summary of results for $\alpha = 1$ and $\alpha = 100$ with Backslash.

1.d

In a situation where it is sufficient to compute a solution to accuracy 10^{-5} , when considering the system with $\alpha = 100$ the GMRES is better than the backslash method, indeed we can see from the table above that GMRES only takes 0.0004 seconds to reach the required level of accuracy (with $m = 5$), while the backslash method takes 0.0030 seconds.

When $\alpha = 1$ we never get the relative residual norm to the magnitude of 10^{-5} with GMRES, so the backslash method would be preferable.

Exercise 2

2.a

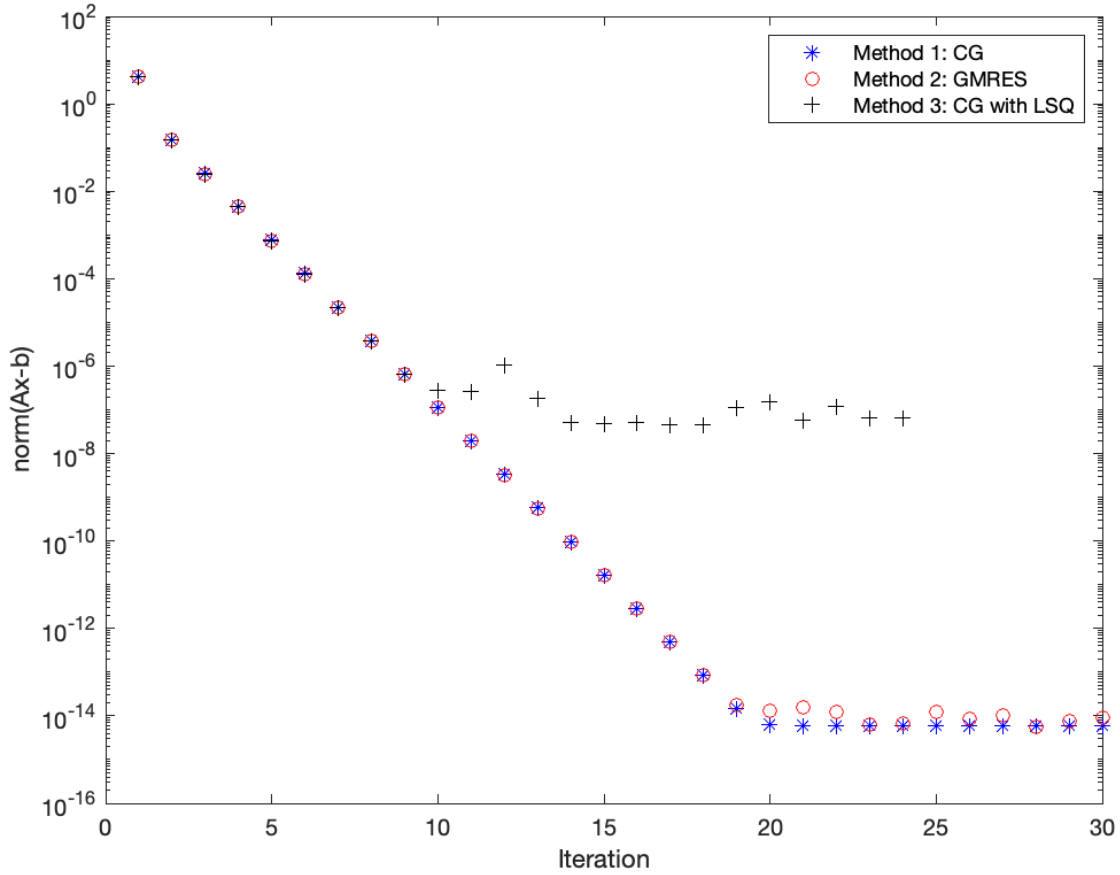


Figure 3: The residual norm $\|Ax - b\|$ with respect to the iterations for the 3 methods

2.b

From the figure, we observe that GMRES and CG achieve nearly identical residual norms at each iteration. However, when zooming in, it becomes evident that CG performs slightly worse in terms of the residual norm compared to GMRES. This difference arises because CG does not include the minimization step present in GMRES.

On the other hand, CG with LSQ initially performs very well, closely matching GMRES. This is because it incorporates a minimization step and, theoretically, the generated x vectors span the Krylov space. In theory, computing a more optimal solution $x = Xz$ by minimizing z over $\|AXz - b\|$, where X represents the span of x vectors generated by CG, is the same as the minimization step in GMRES.

However, in practice as iterations progress, the x vectors generated by CG converge and become less linearly independent as the residual norm decreases, the changes in x become too small. This reduces the numerical stability of LSQ due to an increase in the condition number. Consequently, LSQ becomes less accurate and stable.

2.c

The best method for this problem is CG, since both GMRES and CG ensure the same level of accuracy (alignment in the figure) but CG is less computationally expensive. We can, in fact, verify with the theory that in order to carry out m steps of GMRES, the accumulated computation time is $\mathcal{O}(nm^2)$ (due to the Gram-Schmidt orthogonalization contained in the Arnoldi method).

On the other hand, in the CG method the accumulated computation-time is linear in the number of iterations: $\mathcal{O}(nm)$.

Exercise 3

3.a video 201

An example of application that requires us to solve a linear system is to find the heat distribution in a CPU socket. To model the heat distribution we can use the steady-state heat equation:

$$-k\nabla^2 u(x, y, z) = q(x, y, z)$$

where $u(x, y, z)$ represents the temperature profile. Using a finite element model with a geometry mesh, we then have to solve a linear system of equations:

$$Ax = b$$

where A is usually sparse (to get a more accurate model, we need a finer discretization, that would lead to a larger, sparse system).

To solve the linear system there are two main kinds of methods: direct and iterative methods. Direct methods usually have a pre-fixed number of operations, their complexity is known and they don't find any "early approximation", it means that the method does not find an approximation until it has completed all iterations. On the other hand, with iterative methods the approximation is improved gradually, each iteration is often cheap to compute and they are suitable for very large problems.

3.b video 215

In this video it is shown how to find a convergence factor for the GMRES method when there is an outlier λ_1 in the eigenvalues of the matrix of the linear system $Ax = b$. Starting from the main convergence theorem for GMRES, which states that: if A is diagonalizable, after m iterations, it holds that:

$$\frac{\|Ax_m - b\|}{\|b\|} \leq \|V\| \|V^{-1}\| \min_{p \in P_m^{(0)}} \max_{j=1, \dots, n} |p(\lambda_j)|$$

the idea is try to find $q \in P_m^{(0)}$ such that $q(\lambda_1)$ is zero (so that the bound given from the previous equation is not affected by the outlier) (a polynomial of the form

$$q(z) = \frac{\lambda_1 - z}{\lambda_1} r(z) \quad r \in P_{m-1}^{(0)}$$

To formalize this idea we use the following **Lemma**:
Suppose $(\lambda_2, \dots, \lambda_n) \in D(\rho, c)$ and $\lambda_1 \notin D(\rho, c)$. Then:

$$\min_{p \in P_m^{(0)}} \max_{j=1, \dots, n} |p(\lambda_j)| \leq \frac{|\lambda_1 - c| + \rho}{|\lambda_1|} \left(\frac{\rho}{|c|} \right)^{m-1}$$

Where ρ, c are respectively the radius and center of the disk D and m is the number of iterations. So, combining the result from the Lemma with the main convergence theorem, we can say that if all the eigenvalues of A are contained in a disk D , except for λ_1 , then it holds:

$$\frac{\|Ax_m - b\|}{\|b\|} \leq \|V\| \|V^{-1}\| \frac{|\lambda_1 - c| + \rho}{|\lambda_1|} \left(\frac{\rho}{|c|} \right)^{m-1}$$

3.c video 216

In certain situations GMRES is not efficient, sometimes it does not even converge, because of the eigenvalues of A . The idea is to transform the original system $Ax = b$ multiplying both left and right handside with the inverse of a preconditioner matrix M such that the new system $M^{-1}Ax = M^{-1}b$ is solvable by GMRES.

The preconditioner should be chosen in a way that GMRES converges fast for the new system and it is cheap to compute $M^{-1}b$, in particular we would like to find a preconditioner such that $M \approx A$ (so $M^{-1}A \approx I$ and we need few iterations to get x) and computing $M^{-1}b$ is cheaper than $A^{-1}b$ (because if it is not cheaper, then it is worth to compute $x = A^{-1}b$ directly, without even using GMRES). Some common choices for M are:

- $M = \text{diag}(A)$, works well if A is diagonally dominant.
- M is an upper or lower triangular matrix, since $M^{-1}b$ can be computed with forward or backward substitution.

3.d video 234

The CG method is used for linear systems with a symmetrical, positive definite matrix, so given a generic system $Ax = b$, it is usually necessary to apply preconditioning (both left and right) to solve it with CG. Let M_1 and M_2 be the left and right preconditioner respectively, so the modified system will be:

$$M_1^{-1}AM_2^{-1}M_2x = M_1^{-1}b$$

and being $B = M_1^{-1}AM_2^{-1}$ the matrix of the system, in order to make it SPD, we need to choose

$$M_2^{-1} = M_1^{-T}$$

because

$$(B = B^T) \implies (M_1^{-1}AM_2^{-1}) = (M_1^{-1}AM_2^{-1})^T = (M_2^{-T}AM_1^{-T}) \implies (M_2^{-1} = M_1^{-T})$$

so, $B = M_2^{-T}AM_2^{-1}$ and to select M_2 , it would be favourable for convergence if

$$M_2^T M_2 \approx A$$

so that $B \approx I$ and the method needs few iterations to converge.

3.e video 235

It is computationally easier to find the preconditioner $M = M_2^T M_2$ as an approximation of A instead of computing its factorization M_2 (in the video M_2 is called R). So, in order to write the algorithm of PCG without involving the M_2 matrix we need to apply a change of variables at the algorithm of standard CG:

- $\hat{x} = M_2^{-1}x \implies \hat{x} = \hat{x}_{m-1} - \alpha_m \hat{p}_{m-1}$
- $\hat{r}_m = M_2^T r_m \implies \hat{r}_m = \hat{r}_{m-1} - \alpha_m A \hat{p}_{m-1}$
- $\hat{p}_m = M_2^{-1}r_{m-1} + \beta p_{m-1} \implies \hat{p}_m = M^{-1}\hat{r}_{m-1} + \beta \hat{p}_{m-1}$

3.f video 250

We want to solve $Ax = b$, but now consider also an adjoint system $A^T \hat{x} = \hat{b}$. BiCG can be derived from the application of PCG to:

$$\begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} = \begin{bmatrix} \hat{b} \\ b \end{bmatrix}$$

with preconditioner $M = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$

Comparing the properties of CG and BiCG we can see that they have different search spaces, different orthogonality properties and BiCG has no minimization property and no minmax convergence bounds. Moreover the condition number of the matrix $\begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix}$ is typically smaller than $k(A^T A)$, which is the condition number of CGN, this can be seen as one of the justifications why BiCG performs much better than CGN in practice, although CGN is better understood theoretically.

3.g video 214

The video analyzes the convergence of the GMRES method. In particular the following corollary of the main convergence theorem for GMRES, gives a convergence factor when the eigenvalues of the matrix A of the linear system are all contained in a disk D and D does not include the origin of the plane.

Corollary:

Suppose $\lambda_1, \dots, \lambda_n \in D(\rho, c)$. Then:

$$\frac{\|Ax_m - b\|}{\|b\|} \leq \|V\| \|V^{-1}\| \left(\frac{\rho}{\|c\|}\right)^m$$

where ρ, c are the radius and center of the disk.

Proof:

Consider the polynomial

$$q(z) = \left(\frac{c-z}{c}\right)^m$$

notice that $q \in P_m^{(0)}$ since

- $q \in P_m$
- $q(0) = 1$

Then:

$$\min_{p \in P_m^{(0)}} \max_{j=1, \dots, n} |p(\lambda_j)| \leq \max_{j=1, \dots, n} |q(\lambda_j)| = \max_{j=1, \dots, n} \left(\frac{|c - \lambda_j|}{|c|} \right)^m \leq \left(\frac{\rho}{|c|} \right)^m$$

Plugging this result in the main convergence theorem bound, we obtain:

$$\frac{\|Ax_m - b\|}{\|b\|} \leq \|V\| \|V^{-1}\| \left(\frac{\rho}{|c|} \right)^m$$

So it is favorable when eigenvalues are far from the origin and they are located close to each other in the plan (in this way the convergence factor $\frac{\rho}{\|c\|}$ will be small, giving fast convergence).

Exercise 4

To prove a bound on how many steps of CG must be carried out in order to reduce the error by a factor 10^7 , it is useful to look at the following theorem:

Theorem:

Let $A \in \mathbb{C}^{n \times n}$ be an invertible and symmetric positive definite matrix. Let $x_m, m = 1, \dots$ be a sequence of iterates generated by the Conjugate Gradient (CG) method. Then, for sufficiently large m , the following inequality holds:

$$\|Ax_m - b\|_{A^{-1}} \leq 2 \left(\left(\frac{\sqrt{\kappa} + 1}{\sqrt{\kappa} - 1} \right)^m + \left(\frac{\sqrt{\kappa} + 1}{\sqrt{\kappa} - 1} \right)^{-m} \right)^{-1} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^m$$

where $\|\cdot\|_{A^{-1}}$ denotes the norm induced by the matrix A^{-1} , and κ is the condition number of A with respect to the Euclidean norm.

Moreover, since A is a symmetric positive definite matrix (all eigenvalues are positive by hypothesis), its condition number k can be computed as

$$k = \frac{\lambda_{max}}{\lambda_{min}} \approx \frac{10.5}{2} = 5.25$$

where $\lambda_{max} = 10.5$ and it is reasonable to assume $\lambda_{min} = 2$ since the 100 lowest eigenvalues are all distributed in the interval $I = [2, 3]$.

So, putting the value of the bound

$$2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^m = 10^{-7}$$

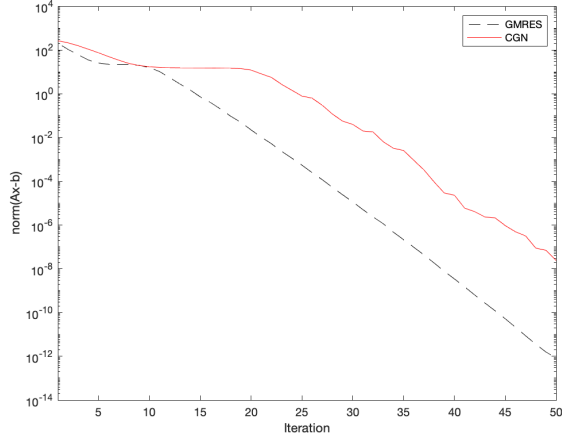
and taking the logarithm of both right and left handside of the equation to isolate the number of iterations m :

$$m = \frac{\log(\frac{10^{-7}}{2})}{\log(\frac{\sqrt{k}-1}{\sqrt{k}+1})} \approx 17.97$$

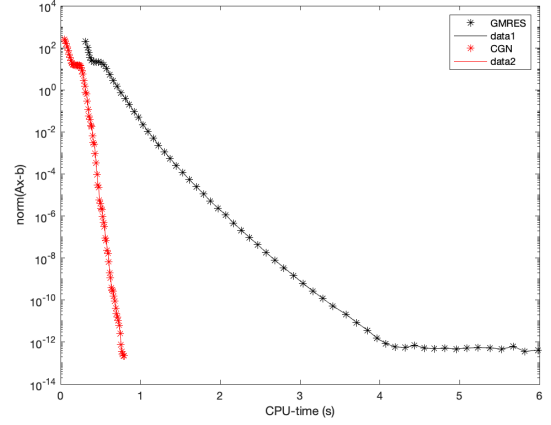
we can conclude that after 18 iterations of the CG, the error will be reduced by a factor 10^7

Exercise 5

5.a



(a) GMRES vs CGN residual norm error with respect to iterations



(b) GMRES vs CGN residual norm error with respect to CPU-time

5.b

In the first figure we notice that CGN converges slightly slower than GMRES. From the theory, we know that both CGN and GMRES methods minimize the residual with respect to the standard Euclidean norm (we can indeed show for CGN that:

$$\|A^T Ax - A^T b\|_{(A^T A)^{-1}}^2 = \|Ax - b\|_2^2$$

But the difference is in the subspace in which the methods search the approximations:

- GMRES minimizes the residual over $K_m(A, b)$
- CGN minimizes the residual over $K_m(A^T A, A^T b)$

These subspaces have different approximation properties and most of the time CGN subspace has worse approximation properties, so our observations are justified by the theory.

In the second figure, we see that CG converges much faster than GMRES, this reflects the results found in the convergence theory of the two methods. We know that CG is optimized for symmetric positive definite systems and it is a three-term recurrence method, meaning that at any point in the execution of the algorithm, only three vectors need to be stored, leading to a linear accumulated computation-time in the iteration count.

Alternatively, the computation time associated with the Gram-Schmidt orthogonalization, contained in GMRES, grows with the iterations, leading to a quadratic accumulated computation-time in the iteration count.